
CY 342: Understanding Small Systems

Course Code	CY 342
Course Name	Understanding Small Systems
L-T-P-C	3-0-0-3
Prerequisites	Consent of the faculty member
Intended for	B. Tech. 3rd Year Semester: Odd/Even: Even
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Big picture and principles of the small world.** [6 Lectures]
- **Why the smaller, the better?** [3 Lectures]
- **Introduction to Nanoscale physics, Nanomaterials.** [12 Lectures]
- **Nanomechanics, Nanoelectronics, Nanophotonics.** [9 Lectures]
- **Nanoscale Fluid Mechanics, Nanoscale Heat transfer.** [6 Lectures]
- **Nanobiology, Molecular motors, future Nanoscience.** [6 Lectures]

Text and Reference Books

1. Masaru Kuno, **Introductory Nanoscience**, Garland Science, 2011.
2. A. N. Clealand, **Foundations of Nanomechanics**, Springer, 2003.
3. Vladimir V. Mitin, Dmitry I. Sementsov & Nizami Z. Vagidov, **Quantum Mechanics for Nanostructures**, Cambridge, 2010.
4. Edward L. Wolf, **Nanophysics and Nanotechnology: An introduction to the modern concepts in Nanoscience**, Wiley-VCH, 2011.